

Thank you for sharing the image. Here's a breakdown and solution approach for each of the questions from your "Workout" sheet:

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1. Line through two points (-2,1) and (4,3)

A. Slope (m):

$$m = \frac{3 - 1}{4 - (-2)} = \frac{2}{6} = \frac{1}{3}$$

B. y-intercept (b):

Use the point-slope form:

$$y - y_1 = m(x - x_1) \rightarrow y - 1 = \frac{1}{3}(x + 2) \rightarrow y = \frac{1}{3}x + \frac{5}{3}$$

C. x-intercept:

Set :

$$0 = \frac{1}{3}x + \frac{5}{3} \rightarrow x = -5$$

D. Equation of the line:

$$y = \frac{1}{3}x + \frac{5}{3}$$

E. Graph:

Plot points (-2,1) and (4,3), draw a straight line through them. Include x-intercept (-5,0) and y-intercept (0,5/3).

## 2. Cost, Revenue, and Profit

Fixed cost = 1200 birr/month

Variable cost = 2.5 birr/unit

Selling price = 10 birr/unit

A. Total Cost (TC):

$$TC = 1200 + 2.5x$$

B. Total Revenue (TR):

$$TR = 10x$$

C. Total Profit (P):

$$P = TR - TC = 10x - (1200 + 2.5x) = 7.5x - 1200$$

D. Break-even point:

$$\begin{aligned} TR &= TC \rightarrow 10x = 1200 + 2.5x \rightarrow \\ 7.5x &= 1200 \rightarrow x = 160 \text{ units} \end{aligned}$$

E. Graph:

Plot TC and TR on the same graph, with x-axis = units sold, y-axis = cost/revenue. Show break-even point at (160, 1600).

### 3. Calculator production optimization

| Type | Circuit | Assembly (h) | Cases | Profit |
|------|---------|--------------|-------|--------|
|------|---------|--------------|-------|--------|

|       |   |   |   |        |
|-------|---|---|---|--------|
| Small | 5 | 1 | 1 | 6 birr |
|-------|---|---|---|--------|

|        |   |   |   |         |
|--------|---|---|---|---------|
| Medium | 7 | 3 | 1 | 13 birr |
|--------|---|---|---|---------|

|       |    |   |   |         |
|-------|----|---|---|---------|
| Large | 10 | 4 | 1 | 20 birr |
|-------|----|---|---|---------|

Constraints:

Circuits:

Assembly:

Cases:

Maximize:

$$\text{Profit} = 6x + 13y + 20z$$

A. Solve this linear programming problem using the Simplex method or graphing (if simplified).

B. Maximum profit is the objective value at optimal point.

## 4. Customer transition (Markov Chains)

Transition matrix:

$$P = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix}$$

Initial state vector:

$$v_0 = \begin{bmatrix} 0.7 \\ 0.3 \end{bmatrix}$$

A.

Week 1:

$$\begin{aligned} v_1 &= P \cdot v_0 = \\ &\begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix} \\ &\cdot \\ &\begin{bmatrix} 0.7 \\ 0.3 \end{bmatrix} \\ &= \\ &\begin{bmatrix} 0.65 \\ 0.35 \end{bmatrix} \end{aligned}$$

Repeat for next weeks using

B.

Long run: Find steady state vector such that .

Solve system:

$$0.8x + 0.3y = x \quad \backslash \backslash$$

$$0.2x + 0.7y = y \quad \backslash \backslash$$

$$x + y = 1$$

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Would you like step-by-step calculation for question 3's optimization or question 4's long-run probabilities?

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Question 4: Customer Switching Between Stores

Let:

: Proportion of customers at store 1 in week

: Proportion of customers at store 2 in week

We are told:

Customers transition weekly as:

From store 1: 80% stay, 20% go to store 2

From store 2: 30% switch to store 1, 70% stay

This gives the transition matrix:

$$\begin{bmatrix} x_{n+1} \\ y_{n+1} \end{bmatrix} = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix} \begin{bmatrix} x_n \\ y_n \end{bmatrix}$$

So:

$$\begin{aligned} x_{n+1} &= 0.8x_n + 0.3y_n \\ y_{n+1} &= 0.2x_n + 0.7y_n \end{aligned}$$

A. Next Week Proportions (Week 1)

$$\begin{aligned} x_1 &= 0.8(0.7) + 0.3(0.3) = 0.56 + 0.09 = 0.65 \\ y_1 &= 0.2(0.7) + 0.7(0.3) = 0.14 + 0.21 = 0.35 \end{aligned}$$

## B. Long Run Proportions

In steady state, , , so:

$$x = 0.8x + 0.3y \rightarrow 0.2x = 0.3y \rightarrow$$

$$x = \frac{3}{2}y$$

$$x + y = 1 \rightarrow \frac{3}{2}y + y = 1$$

$$\rightarrow \frac{5}{2}y = 1 \rightarrow y =$$

$$\frac{2}{5} = 0.4 \rightarrow x = 0.6$$

So in the long run:

Store 1: 60%

Store 2: 40%

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## Question 5: Linear Programming for Car Production

Let:

: number of Abay cars

: number of Tekeze cars

Constraints:

Assembly:

Assembly:

Packaging:

Engines:

Non-negativity:

Objective function:

Maximize profit:

A. Linear Programming Model:

$\text{Maximize } Z = 65x + 90y$

$\text{Subject to:}$

$10x + 14y \leq 720$

$2x + 4y \leq 300$

$x + y \leq 75$

$x \geq 0, y \geq 0$

B & C. Solve Using Simplex Method

I can solve this step-by-step if you'd like, or graphically for a quick answer.

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Would you like a detailed solution using the simplex method, or a graphical method to find the

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Here's how to approach the questions in your image step-by-step:

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### Q1: Journal Entries for December Transactions

Date Account Titles and Explanation Debit (Birr)  
Credit (Birr)

Dec 1 Cash 15,000

Owner's Capital 15,000

Dec 2 Rent Expense 2,400

Cash 2,400

Dec 3 Automobile Expense 750

Miscellaneous Expense 380

Cash 1,130

Dec 4 Supplies 950

Accounts Payable 950

Dec 5 Cash 17,350

Commission Revenue 17,350

Dec 6 Accounts Payable 580

Cash 580

Dec 7 Salaries Expense 3,600

Cash 3,600

Dec 8 Drawings 1,500

Cash 1,500

Dec 9 Supplies Expense 675

Dec 9 Supplies Expense 675

Supplies 675

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Q1B: Financial Statements as of December 31,  
2015

Income Statement

Revenues

Commission Revenue: 17,350

Expenses

Rent Expense: 2,400

Automobile Expense: 750

Miscellaneous Expense: 380

Salaries Expense: 3,600

Supplies Expense: 675

Total Expenses: 7,805

Net Income =  $17,350 - 7,805 = 9,545$

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## **Owner's Equity Statement**

**Beginning Capital: 0**

**Add: Investment: 15,000**

**Add: Net Income: 9,545**

**Less: Drawings: 1,500**

**Ending Capital: 23,045**

## **Balance Sheet**

### **Assets**

**Cash:  $15,000 - 2,400 - 1,130 + 17,350 - 580 - 3,600 -$**